User Lifecycle and Provisioning

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Automating User Management at Scale

Manual user management doesn't scale. This notebook covers user lifecycle automation including SCIM provisioning, JIT access, service accounts, and token management.

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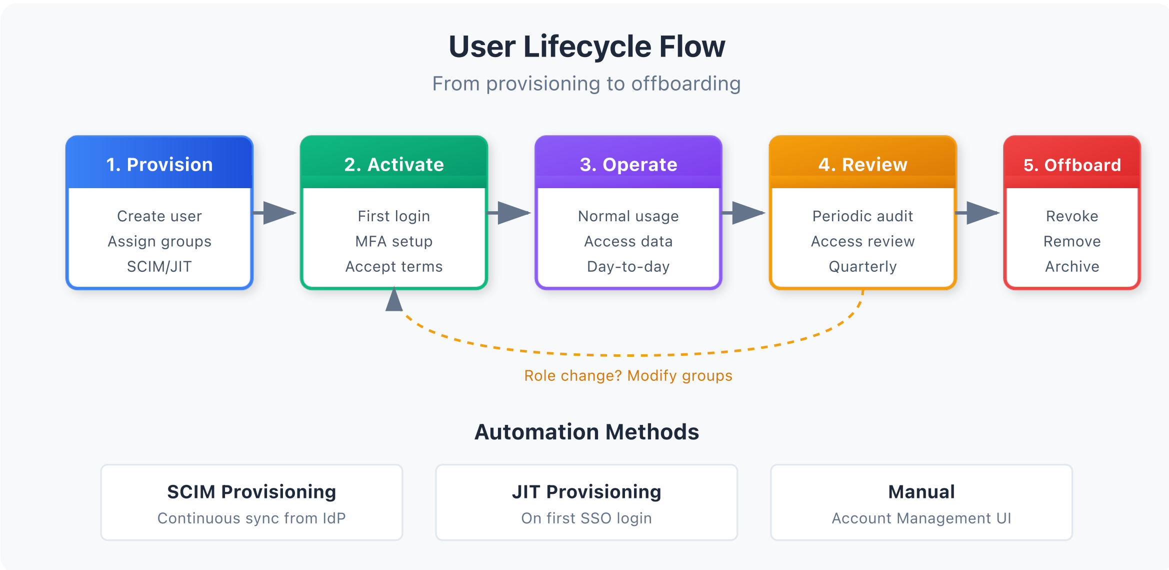
Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Gen3 IAM enabled
Permissions	`account-iam-admin` for user/token management
IdP	Enterprise IdP (Okta, Azure AD, etc.) for SCIM

1. User Lifecycle Overview

Users have a lifecycle from creation to deactivation. Effective management requires automation at each stage.



Lifecycle Stages

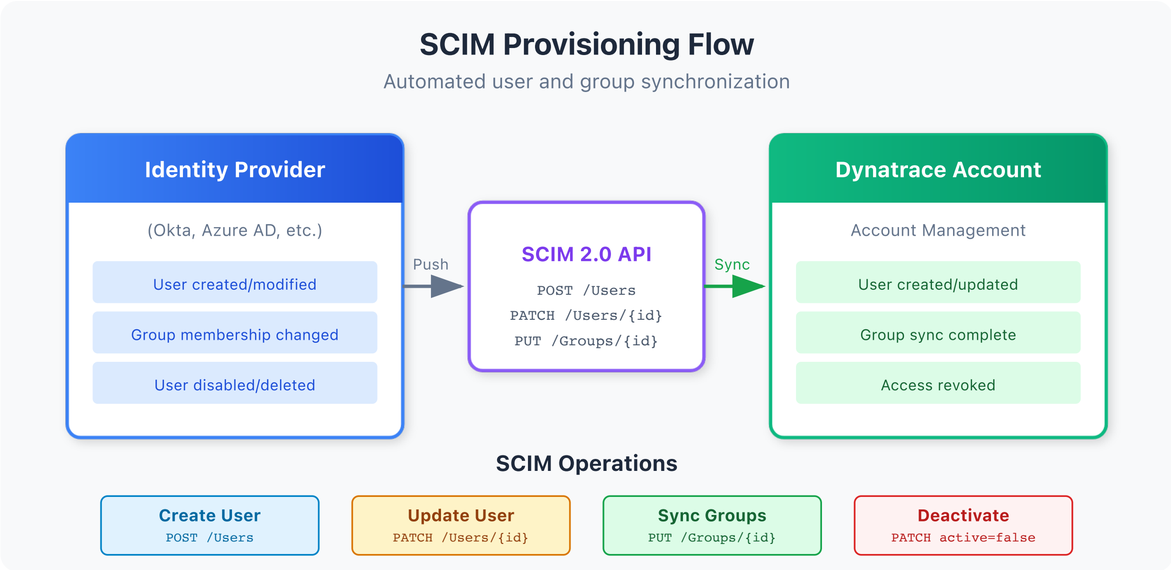
Stage	Description	Automation
Provision	Create user identity	SCIM, JIT
Assign	Add to groups, grant access	SCIM group sync
Activate	First login, MFA setup	SSO integration
Operate	Normal access and usage	N/A
Review	Periodic access review	Audit queries
Modify	Change groups/access	SCIM update
Offboard	Remove access	SCIM delete, JIT expiry

Manual vs Automated

Method	Pros	Cons	Scale
Manual	Full control	Slow, error-prone	< 50 users
SCIM	Automated sync	Setup complexity	50 - 10,000+ users
JIT	No pre-provisioning	Less control	Any size

2. SCIM Provisioning

SCIM (System for Cross-domain Identity Management) automatically syncs users and groups from your IdP to Dynatrace.



How SCIM Works

1. You configure SCIM in your IdP (Okta, Azure AD, etc.)
2. When users/groups change in IdP, SCIM syncs to Dynatrace
3. Users are automatically created, updated, or deactivated
4. Group memberships stay in sync

Supported IdPs

IdP	SCIM Support	Notes
Okta	Full	Native integration
Azure AD (Entra ID)	Full	Enterprise app provisioning
OneLogin	Full	Directory provisioning
PingFederate	Full	Provisioning connector
Google Workspace	Partial	Requires adapter

SCIM Configuration Steps

In Dynatrace:

1. Navigate to Account Management → Identity providers
2. Select your IdP configuration

3. Enable SCIM provisioning
4. Copy the SCIM endpoint URL and token

In Your IdP (Example: Okta):

1. Open your Dynatrace SAML application
2. Go to Provisioning tab
3. Configure SCIM connection:
   - SCIM connector base URL
   - API token (from Dynatrace)
4. Enable provisioning features:
   - Create Users
   - Update User Attributes
   - Deactivate Users
   - Push Groups

SCIM Attributes Synced

Attribute	Source	Dynatrace Field
userName	IdP email	Email (login)
givenName	IdP first name	First name
familyName	IdP last name	Last name
active	IdP status	Account status
groups	IdP group membership	Group membership

Best Practices

- **Test in sandbox** before production
- **Map groups explicitly** - don't auto-create
- **Use consistent naming** between IdP and Dynatrace
- **Monitor sync status** for failures

3. Just-in-Time Provisioning

JIT provisioning creates users automatically on first login via SSO.

How JIT Works

1. User authenticates via SAML SSO
2. If user doesn't exist, Dynatrace creates them
3. Group memberships assigned based on SAML assertion
4. User accesses Dynatrace immediately

JIT vs SCIM

Aspect	JIT	SCIM
User created	On first login	Before login
Deactivation	Manual or session-based	Automatic sync
Group sync	At login only	Continuous
Setup	Simple	More complex
Control	Less	More

Configuring JIT

1. Navigate to Account Management → Identity providers
2. Select your SAML configuration
3. Enable "Auto-provision users on login"
4. Configure default group for new users
5. Map SAML groups to Dynatrace groups

JIT Group Mapping

SAML assertion includes groups, which are mapped to Dynatrace groups:

SAML Group	Dynatrace Group
`IT-Dynatrace-Admins`	`dt-platform-admins`
`App-Checkout-Team`	`dt-checkout-editors`
`All-Employees`	`dt-all-viewers`

JIT Considerations

Pro	Con
Simple setup	User created only after login
No pre-provisioning needed	Group sync only at login
Self-service access	Harder to audit who has access

4. User Onboarding Workflows

Standard Onboarding Process

Step	Action	Owner
1	Add user to IdP group	HR/IT
2	SCIM syncs user to Dynatrace	Automatic
3	User receives welcome email	IT
4	User logs in via SSO	User
5	User completes MFA setup	User
6	Access verified	User/Manager

New Employee Checklist

```
□ Employee added to corporate directory
□ Employee assigned to appropriate IdP groups
□ SCIM sync verified (user appears in Dynatrace)
□ Employee logs in successfully
□ Employee completes MFA enrollment
□ Employee confirms access to required data
```

Role Change Process

When users change roles:

1. Manager requests access change
2. IAM admin approves
3. Update IdP group membership
4. SCIM syncs changes
5. User has new access on next login

Temporary Access

For contractors or short-term access:

Method	Duration	Automation
IdP time-limited group	Days to months	IdP policy
Manual group removal	Any	Calendar reminder
OAuth client expiry	Hours to days	Token config

5. User Offboarding

Proper offboarding is critical for security.

Offboarding Triggers

Trigger	Response	Timing
Termination	Full revocation	Immediate
Resignation	Full revocation	Last day
Role change	Partial revocation	At change
Contract end	Full revocation	End date

Automated Offboarding (SCIM)

1. User disabled/removed in IdP
2. SCIM syncs deactivation to Dynatrace
3. User cannot login
4. Existing sessions invalidated

Manual Offboarding Checklist

```
□ Remove from all Dynatrace groups
□ Revoke any API tokens created by user
□ Deactivate OAuth clients owned by user
□ Transfer ownership of dashboards/notebooks
□ Review audit log for recent activity
□ Document offboarding completion
```

Token Revocation on Offboarding

Critical: User-created tokens remain active until explicitly revoked.

Query to find user's tokens:

```
Account Management → API tokens
Filter by: Created by [user email]
Action: Revoke all tokens
```

Verification

After offboarding, verify:

- [ ] User cannot login via SSO
- [ ] User not in any groups
- [ ] User's tokens are revoked
- [ ] No active OAuth clients

6. Service Accounts and OAuth Clients

Non-human identities require special handling.

Service Accounts vs OAuth Clients

Type	Use Case	Authentication
Service Account	Automation, scripts	API token
OAuth Client	Apps, integrations	OAuth 2.0

Creating OAuth Clients

1. Account Management → OAuth clients
2. Create new client
3. Configure:
   - Name and description
   - Grant type (client credentials)
   - Scopes (permissions)
   - Token lifetime
4. Save client ID and secret securely

OAuth Client Best Practices

Practice	Rationale
One client per application	Isolation, easy revocation
Minimum scopes	Least privilege
Short token lifetime	Reduce exposure
Rotate secrets regularly	Security hygiene
Document ownership	Accountability

OAuth Scopes

Common scopes for automation:

Scope	Grants
`storage:logs:read`	Read logs
`storage:metrics:read`	Read metrics
`settings:objects:read`	Read settings
`settings:objects:write`	Modify settings
`document:documents:read`	Read dashboards

Service Account Lifecycle

Event	Action
Create	Document owner, purpose, expiry
Review (quarterly)	Verify still needed
Owner change	Update documentation
Decommission	Revoke client, archive docs

7. API Token Management

API tokens provide access for scripts, tools, and legacy integrations.

Token Types

Type	Scope	Use Case
Personal	User's permissions	User automation
Service	Specific scopes	Application integration

Token Scopes Reference

Scope	Permission
`logs.read`	Read log data
`metrics.read`	Read metrics
`entities.read`	Read entity data
`settings.read`	Read settings
`settings.write`	Modify settings
`problems.read`	Read problems
`events.ingest`	Send events

Token Best Practices

Practice	Implementation
Minimum scopes	Only grant what's needed
Descriptive names	`cicd-pipeline-metrics-reader`
Expiration dates	Set reasonable expiry
Rotation schedule	Rotate every 90 days
Secure storage	Use secrets manager
No sharing	One token per use case

Token Rotation Process

1. Create new token with same scopes
2. Update applications with new token
3. Verify applications work
4. Revoke old token
5. Document rotation

Monitoring Token Usage

Use audit logs to track token activity.

```
// Track API token usage patterns
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "token")
| fields timestamp, content
| sort timestamp desc
| limit 100
```

```
// Find token creation events
fetch logs, from: now() - 30d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "token") and matchesPhrase(content,
"created")
| fields timestamp, content
| sort timestamp desc
| limit 50
```

```
// Monitor OAuth client activity
fetch logs, from: now() - 7d
| filter matchesPhrase(log.source, "audit")
| filter matchesPhrase(content, "oauth") or matchesPhrase(content, "client")
| fields timestamp, content
| sort timestamp desc
| limit 50
```

Next Steps

With user lifecycle management in place, proceed to compliance:

Recommended Path

1. **IAMADM-07: Audit Logging and Compliance** - Monitor and report on access
2. **IAMADM-08: Multi-Environment IAM** - Scale across environments
3. **IAMADM-09: Troubleshooting Access Issues** - Debug problems

User Lifecycle Checklist

Before moving on, ensure you have:

- [ ] Chosen SCIM, JIT, or manual provisioning
- [ ] Configured IdP integration for provisioning
- [ ] Documented onboarding process
- [ ] Documented offboarding process
- [ ] Established token management policy
- [ ] Created service account inventory

Summary

In this notebook, you learned:

- User lifecycle stages and automation options
- SCIM provisioning configuration
- Just-in-time provisioning via SSO
- User onboarding and offboarding workflows
- Service account and OAuth client management
- API token best practices and rotation

References

- SCIM Provisioning
- OAuth Clients
- API Tokens
